PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	SONI
Program/Batch	B084-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (Footprinting and reconnaissance with the HARVESTER) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.0.4** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified ten live hosts:

- 192.168.1.1
- 192.168.1.2
- 192.168.1.3
- 192.168.1.4
- 192.168.1.7
- 192.168.1.8
- 192.168.1.33
- 192.168.1.54
- 192.168.1.64

- 192.168.1.56

The results also included ten MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.168.1.56	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.

3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.

5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.

6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.

7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.

8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.

9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

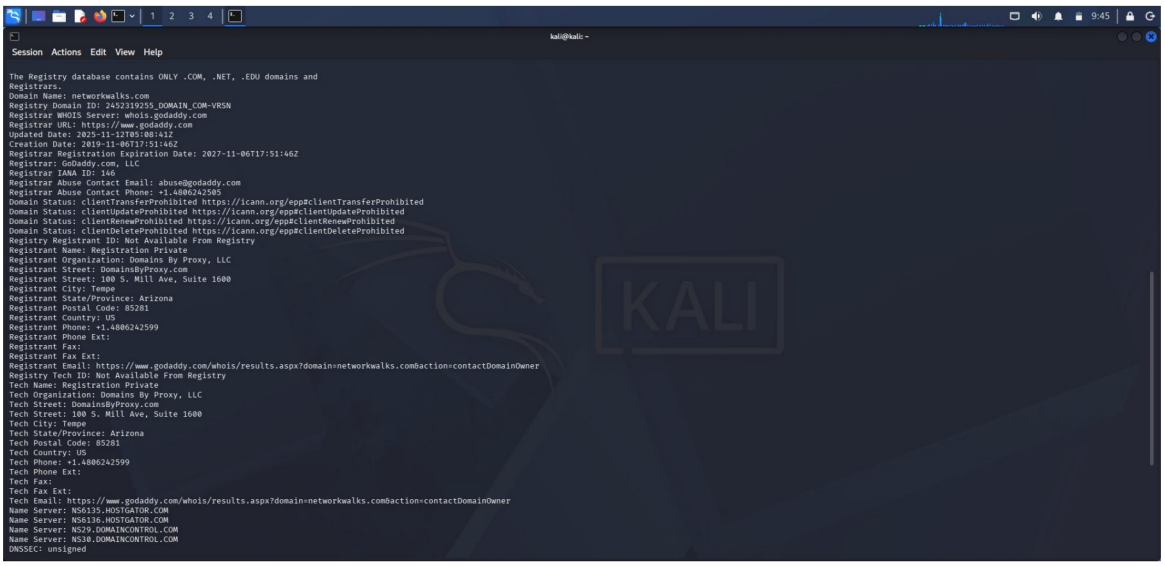
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

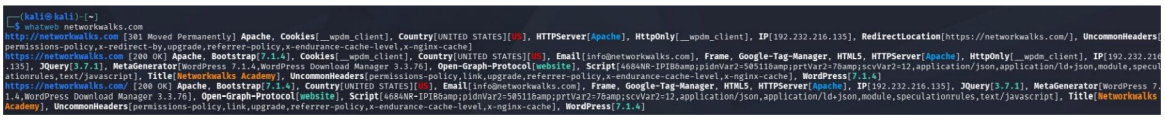
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

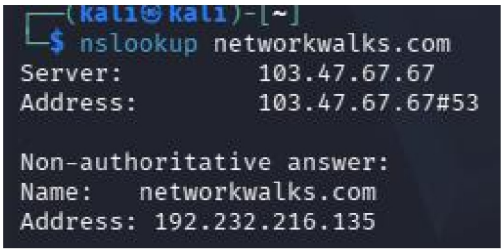
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected


kali@kali: ~
Session Actions Edit View Help
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM
Name Server: NS30.DOMAINCONTROL.COM
DNSSEC: unsigned


(kali㉿kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684#NR-IPI80&pidnVar2=505110&prtVar2=70&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684#NR-IPI80&pidnVar2=505110&prtVar2=70&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]


(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server: 103.47.67.67
Address: 103.47.67.67#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135

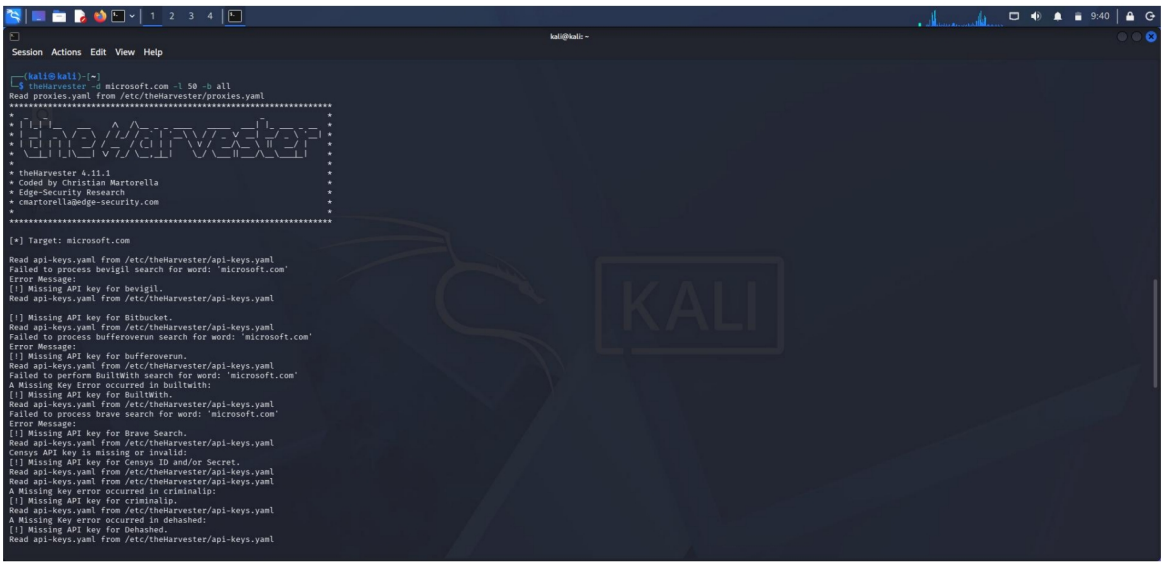
```
┌──(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com

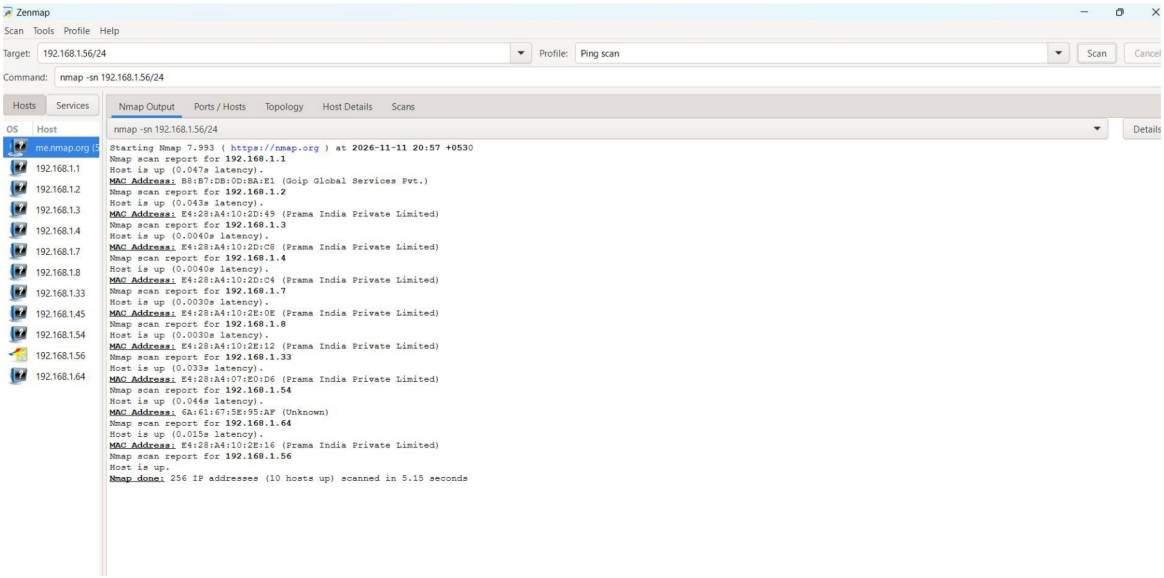
                ?                 ,.   (   .      )       .      "
      __          ??            ("     )  )'     ,'        )  . (`     ``
 (___()'`;   ???              .; )  ' (( (" )    ;(,     ((  (  ;)  "  )")
 /,___ /`                    _"., ,._'_.,)_(..,( . )_  _' )_') (. _..( ' )
 \\   \\                    |___|___|___|___|___|___|___|___|___|___|

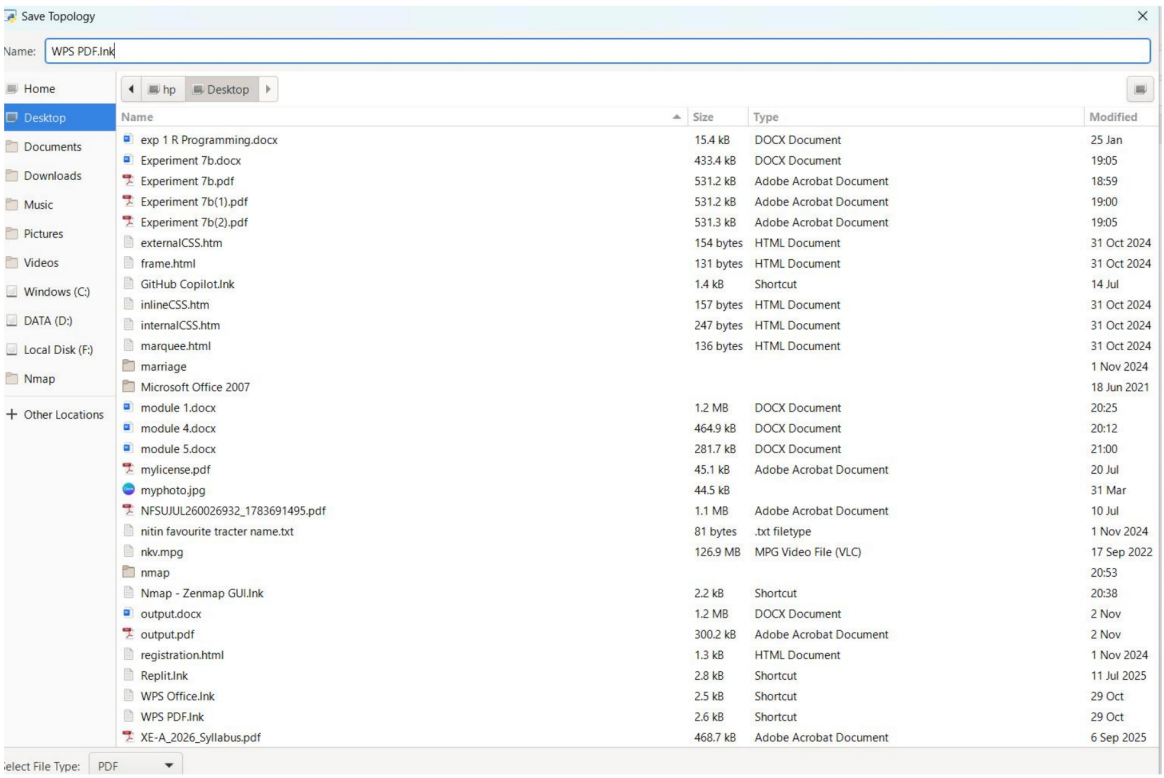
                          ~ WAFW00F : v2.4.2 ~
                ~ Sniffing Web Application Firewalls since 2009 ~

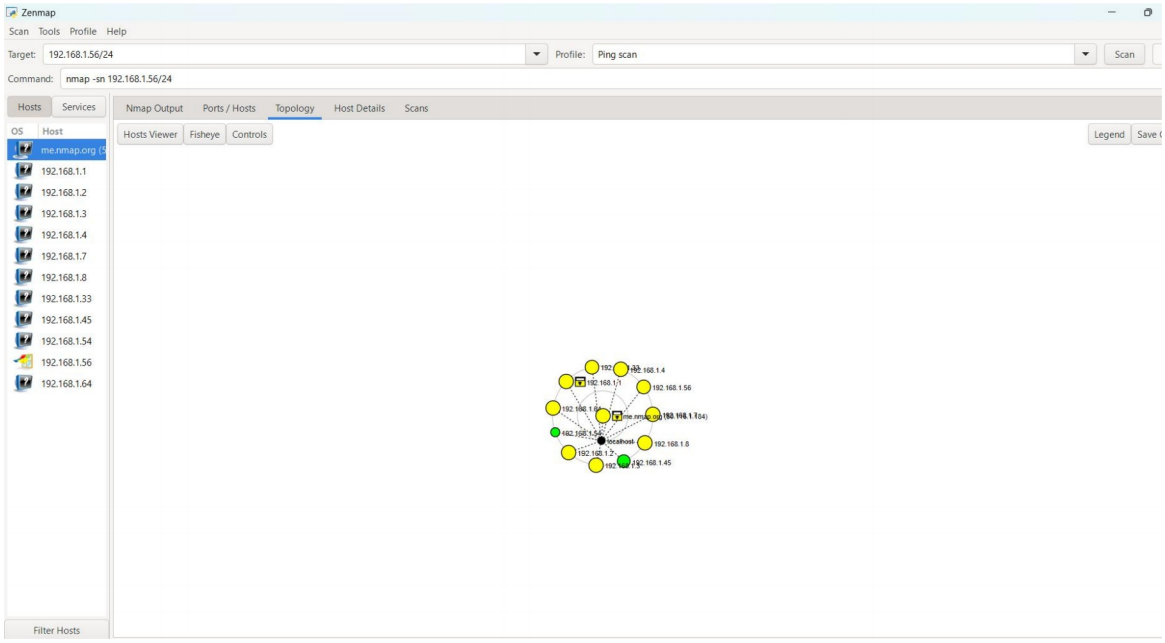
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

```
*******************************************************************
*  _   _                                            _             *
* | |_| |__   ___    ^  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| V /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 4.11.1                                             *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************
```


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
[*] Target: microsoft.com
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml


Zenmap
Scan Tools Profile Help
Target: 192.168.1.56/24
Profile: Ping scan
Scan
Command: nmap -sn 192.168.1.56/24
Hosts Services
OS Host
me.nmap.org
192.168.1.1
192.168.1.2
192.168.1.3
192.168.1.4
192.168.1.7
192.168.1.8
192.168.1.33
192.168.1.45
192.168.1.54
192.168.1.56
192.168.1.64
Nmap Output Ports / Hosts Topology Host Details Scans
nmap -sn 192.168.1.56/24
Starting Nmap 7.993 (https://nmap.org) at 2026-11-11 20:57 +0530
Nmap scan report for 192.168.1.1
Host is up (0.047s latency).
MAC Address: B8:B7:DB:0D:BA:E1 (Goip Global Services Pvt.)
Nmap scan report for 192.168.1.2
Host is up (0.043s latency).
MAC Address: E4:28:A4:10:2D:49 (Prama India Private Limited)
Nmap scan report for 192.168.1.3
Host is up (0.0040s latency).
MAC Address: E4:28:A4:10:2D:C8 (Prama India Private Limited)
Nmap scan report for 192.168.1.4
Host is up (0.0040s latency).
MAC Address: E4:28:A4:10:2D:C4 (Prama India Private Limited)
Nmap scan report for 192.168.1.7
Host is up (0.0030s latency).
MAC Address: E4:28:A4:10:2E:0E (Prama India Private Limited)
Nmap scan report for 192.168.1.8
Host is up (0.0030s latency).
MAC Address: E4:28:A4:10:2E:12 (Prama India Private Limited)
Nmap scan report for 192.168.1.33
Host is up (0.033s latency).
MAC Address: E4:28:A4:07:E0:D6 (Prama India Private Limited)
Nmap scan report for 192.168.1.54
Host is up (0.044s latency).
MAC Address: 6A:61:67:5E:95:AF (Unknown)
Nmap scan report for 192.168.1.64
Host is up (0.015s latency).
MAC Address: E4:28:A4:10:2E:16 (Prama India Private Limited)
Nmap scan report for 192.168.1.56
Host is up.
Nmap done: 256 IP addresses (10 hosts up) scanned in 5.15 seconds


Save Topology
Name: WPS PDF.lnk
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
DATA (D:)
Local Disk (F:)
Nmap
Other Locations
hp
Desktop
Name
Size
Type
Modified
exp 1 R Programming.docx 15.4 kB DOCX Document 25 Jan
Experiment 7b.docx 433.4 kB DOCX Document 19:05
Experiment 7b.pdf 531.2 kB Adobe Acrobat Document 18:59
Experiment 7b(1).pdf 531.2 kB Adobe Acrobat Document 19:00
Experiment 7b(2).pdf 531.3 kB Adobe Acrobat Document 19:05
externalCSS.htm 154 bytes HTML Document 31 Oct 2024
frame.html 131 bytes HTML Document 31 Oct 2024
GitHub Copilot.lnk 1.4 kB Shortcut 14 Jul
inlineCSS.htm 157 bytes HTML Document 31 Oct 2024
internalCSS.htm 247 bytes HTML Document 31 Oct 2024
marquee.html 136 bytes HTML Document 31 Oct 2024
marriage 1 Nov 2024
Microsoft Office 2007 18 Jun 2021
module 1.docx 1.2 MB DOCX Document 20:25
module 4.docx 464.9 kB DOCX Document 20:12
module 5.docx 281.7 kB DOCX Document 21:00
mylicense.pdf 45.1 kB Adobe Acrobat Document 20 Jul
myphoto.jpg 44.5 kB 31 Mar
NFSUJUL260026932_1783691495.pdf 1.1 MB Adobe Acrobat Document 10 Jul
nitin favourite tracter name.txt 81 bytes .txt filetype 1 Nov 2024
nkv.mpg 126.9 MB MPG Video File (VLC) 17 Sep 2022
nmap 20:53
Nmap - Zenmap GUI.lnk 2.2 kB Shortcut 20:38
output.docx 1.2 MB DOCX Document 2 Nov
output.pdf 300.2 kB Adobe Acrobat Document 2 Nov
registration.html 1.3 kB HTML Document 1 Nov 2024
Replit.lnk 2.8 kB Shortcut 11 Jul 2025
WPS Office.lnk 2.5 kB Shortcut 29 Oct
WPS PDF.lnk 2.6 kB Shortcut 29 Oct
XE-A_2026_Syllabus.pdf 468.7 kB Adobe Acrobat Document 6 Sep 2025
elect File Type: PDF


Zenmap
Scan Tools Profile Help
Target: 192.168.1.56/24
Profile: Ping scan
Scan
Command: nmap -sn 192.168.1.56/24
Hosts
Services
OS Host
me.nmap.org
192.168.1.1
192.168.1.2
192.168.1.3
192.168.1.4
192.168.1.7
192.168.1.8
192.168.1.33
192.168.1.45
192.168.1.54
192.168.1.56
192.168.1.64
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
192.168.1.4
192.168.1.1
192.168.1.56
localhost
192.168.1.8
192.168.1.45

-End-

Author

SONI
Cybersecurity Professional B084

Project Information

Program Name: Cybersecurity program at Networkwalks | Week: 02 | Repository: GitHub